Practice Quiz: Planning

Part A: Multiple Choice

1. The number of policies in a flat POMDP with K actions and planning horizon T is: A) K + T B) K · T C) K^T — exponential in the planning horizon D) T^K
2. Sophisticated inference extends basic EFE by: A) Adding more hidden states B) Simulating future observations and the belief updates they will produce — recursive meta-cognition C) Removing epistemic value D) Using only pragmatic considerations
3. In a hierarchical POMDP, higher levels: A) Process data at a faster timescale B) Set context (initial states, preferences) for lower levels and operate at slower timescales C) Are identical to lower levels D) Only handle sensory processing
4. The computational explosion of planning can be managed by: A) Ignoring future consequences entirely B) Policy pruning, habit formation, and hierarchical decomposition C) Always planning to the maximum depth D) Using random policy selection
5. Deep temporal planning requires: A) Only current observations B) Predicting future states, observations, and belief updates over the planning horizon C) Complete knowledge of the environment D) No generative model
6. A habit in Active Inference terms is: A) A random behavior pattern B) A policy with consistently low EFE that can be selected without full EFE evaluation — reducing computational cost C) An innate reflex D) A policy forced by external constraints
7. The relationship between Active Inference planning and dynamic programming (Bellman equation) is: A) They are identical B) Active Inference uses EFE (which includes epistemic value) instead of expected reward, and evaluates policies rather than value functions C) Active Inference does not involve future planning D) Dynamic programming includes exploration automatically

Part B: Short Answer

1. For a 2-action, 3-step POMDP, enumerate all $2^3 = 8$ policies. For one specific policy, show the step-by-step computation of predicted states, predicted observations, and the EFE at each timestep.
2. Explain how sophisticated inference enables the agent to "change its mind" mid-plan. Why is this important for adaptive behavior? Use a concrete example (e.g., an agent navigating a maze that discovers a blocked path).
3. Design a two-level hierarchical POMDP for a restaurant scenario: Level 2 selects between "appetizer," "main course," and "dessert" sub-goals; Level 1 handles individual actions within each sub-goal. Describe the A, B, C, D matrices at each level and explain how Level 2 sets the context for Level 1.